AUTO-SCALING

1. 1-What is an EC2 instance type, and how do you choose the right one for your application?
2. 2-What is an EC2 instance family, and when would you use one family over another?
3. 3-Describe the typical steps involved in launching an EC2 instance.
4. 4-What is an EC2 user data script, and how can it be used during instance launch?
5. 5-Explain the purpose of EC2 instance metadata and how you can access it from within an instance.
6. 6-How can you create custom AMIs, and why might you want to do so?
7. 7-What are security groups, and how do they control inbound and outbound traffic to EC2 instances?
8. 8-Explain the use of Network Access Control Lists (NACLs) and how they differ from security groups.
9. 9-How do you enable and configure AWS Web Application Firewall (WAF) in front of an EC2-based web application?
10. 10-What is Auto Scaling, and how can it be used to ensure high availability and scalability of EC2 instances?
11. 11-Explain the purpose of Amazon Elastic Load Balancing (ELB) and its integration with EC2 instances.
12. 12-What is Amazon EC2 Container Service (ECS), and how does it help with containerized applications on EC2 instances?
13. 13-How can you configure Amazon Route 53 for DNS-based load balancing of EC2 instances?
14. 14-What is status check in EC2 instance?
15. 15-How to changes instance types for running application without downtime of application?

16. 16-What is difference between AMI and Snapshot
17. 17-How to boot related issues like kernal panic in ec2 Instances?
18. 18-How many maximum number of Ips can be attached to a instance?
19. 19-Describe different types of purchasing options available in aws?
20. 20-What are the different types of AWS Placement Groups, and how do they differ?
21. 21-Can you change the placement group of a running EC2 instance?
22. 22-What is the difference between an Availability Zone and a Placement Group?
23. 23-What are some best practices for using Placement Groups in AWS?
24. 24-Explain the limitations or constraints of AWS Placement Groups?
25. 25-Give examples of scenarios or applications where each type of EBS volume (gp2, io1, st1, sc1, gp3) is the most appropriate choice.
26. 26-What is Amazon Elastic Block Store (EBS), and how does it differ from Amazon S3?
27. 27-What are the different types of EBS volumes available, and when would you use each type (e.g., gp2, io1, st1, sc1, gp3)?
28. 28-Explain the concept of Provisioned IOPS (PIOPS) and when it's necessary for achieving consistent performance.
29. 29-How do you resize an EBS volume, and what precautions should be taken when doing so?
30. 30What is the difference between EBS volume types and EBS volume size, and how do they impact performance?
31. 31-What is an EBS snapshot, and why is it important for data durability and disaster recovery?

32. 32-How often should you create EBS snapshots, and what strategies can be employed for efficient backup and retention policies?
33. 33-What are the best practices for encrypting EBS volumes at rest, and how do you implement encryption?
34. 34-Describe the difference between EBS-backed and instance-store-backed EC2 instances and their respective advantages and limitations.
35. 35-How can you monitor the performance and health of EBS volumes, and what AWS services or tools can assist in this process?